AI:

don’t brick your brain!

DM

10 November 2026

AI is changing mathematics

8 September 2026

Navier–Stokes

OpenAI announces a solution to the Millennium Prize problem about fluid motion.

6 October 2026

700+ papers

OpenAI releases AI-generated mathematical manuscripts together in one batch.

Reactions in the press



Tristan Buckmaster



Hugo Duminil-Copin

The Washington Post 13 September 2026

"He was close to a huge math breakthrough. Then he got scooped by AI."

Le Monde 12 October 2026

"They have annihilated my field of research."

September announcement · October release · Newspaper excerpt 1 · Newspaper excerpt 2

“Better” work can hide worse learning

	Practice / homework	Unaided exams
Turkey Nearly 1,000 pupils Randomised trial	+48%	−17%
China 26,811 pupils, nine subjects Working paper	+18%	−20%

Turkey: compared with a no-AI group. Teacher-designed hints largely avoided the harm, without a significant unaided gain.

China: observational estimates within six months of adoption, relative to pre-adoption means.

Bastani et al. (2025), PNAS Strömberg, Lei and Wu (2026), CEPR Discussion Paper 21577

Subgroups of D_4: a bad use

Question: find all subgroups of D_4, the eight symmetries of a square.

Bad prompt

"Find all subgroups of D_4"

Example AI response (abridged)

There are ten subgroups:

Order	Subgroups
1	$\{e\}$
2	$\langle \pi^2 \rangle$, $\langle \sigma \rangle$, $\langle \pi\sigma \rangle$, $\langle \pi^2\sigma \rangle$, $\langle \pi^3\sigma \rangle$
4	$\langle \pi \rangle$, $\langle \pi^2, \sigma \rangle$, $\langle \pi^2, \pi\sigma \rangle$
8	D_4

The relations verify the subgroup property. A case split into rotations and reflections proves completeness.

Copying this leaves the reasoning unpractised. The quick result can reinforce the habit of skipping practice.

Subgroups of D_4: a good interaction

Student
"I have found only cyclic subgroups. Ask one question to help me check completeness. Do not give the missing subgroups."

AI
"Could two commuting elements of order two generate a non-cyclic subgroup?"

Student, after working independently
"Yes. I tried $K = \{e, \pi^2, \sigma, \pi^2\sigma\}$. Give me one hint to check it."

AI
"Since π^2 commutes with σ, what happens when you multiply two elements of K?"

You still prove closure and completeness.

Open the D4 Closure Lab

d4-closure-lab-daniel.buddingslug4.chatgpt.site

A preview before the lecture

Example: vector spaces

“Tomorrow’s lecture introduces vector spaces. Give me a five-minute overview using $\mathbb{R}^2$ and one example beyond arrows. Explain the role of the field. Then ask me two questions. Keep formal proofs for later.”

- Arrive with a general idea and questions of your own.
- Compare the preview with the lecturer’s precise definitions.
- For difficult reading, request a simpler explanation, then create your own example and non-example.

Good, bad and neutral uses

Good	Extra exercises with answers hidden. AI questions you about a concept you explain. **You practise the skill.**
Bad	AI chooses every next step. Summaries replace reading and checking. **You hand over the skill.**
Neutral	Typesetting your own proof. Grammar or timetable help. **You save time on support work.**

Matrix arithmetic is the test case.

Delegating it can save time when the goal is interpreting a result. It removes the practice when arithmetic is what you are learning.

Your independent self-test

A simple routine

Before
Choose the skill. Make an attempt.

During
Request a small hint. Pause and continue yourself.

After
Close the chat. Rebuild the argument from a blank page.

Your independent test

Can I explain each step and its assumptions?

Can I solve a related problem tomorrow?

Am I needing less help over time?

If not, return to definitions or an easier example. Ask a tutor or peer.

Measure independent ability, not pages completed.
Follow the course's AI and disclosure rules.

Research sources and pictures

Newspaper excerpts

Miriam Waldvogel, The Washington Post, 13 September 2026

David Larousserie, Le Monde, 12 October 2026

Research announcements

OpenAI announcement, 8 September 2026

OpenAI mathematics release, 6 October 2026

Primary manuscript catalogue: openai/math

Bastounis, Circelli and Hansen (2026), arXiv:2610.08144

The latter preprint questions whether the formalisation matches the written proof. An announcement and an independent verification differ.

Pictures

Tristan Buckmaster: portrait from his NYU homepage

Hugo Duminil-Copin: Tatjana Ruf / MFO, 2014, via Wikimedia Commons

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Study sources and comparisons

Bastani et al. (2025)
Generative AI without guardrails can harm learning: Evidence from high school mathematics.

PNAS 122(26), e2422633122. Published paper

Open full text

Strömberg, Lei and Wu (2026)
The Generative AI Learning Penalty: Evidence from Chinese Secondary Education.

CEPR Discussion Paper 21577. Working paper

Authors' explanation, VoxEU, 4 September 2026

The percentages are reported relative score changes, not percentage-point differences. The comparisons differ: a no-AI control in Turkey and estimated changes relative to pre-adoption means in China.

The evidence concerns particular educational settings. It does not establish inevitable skill loss for every AI user.